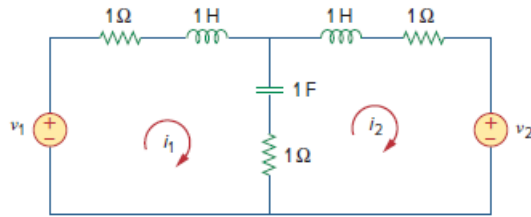


Problem

In the circuit of Fig. 10.76, determine the mesh currents i_1 and i_2 . Let $v_1 = 10 \cos 4t$ V and $v_2 = 20 \cos(4t - 30^\circ)$ V.

Figure 10.76:



Step-by-step solution

Step 1 of 5

Refer to Figure 10.76 in the textbook.

The voltage sources are,

$$v_1 = 10 \cos 4t \text{ V}$$

And,

$$v_2 = 20 \cos(4t - 30^\circ) \text{ V}$$

Convert voltage sources into phasor form.

$$10 \cos 4t \text{ V} = 10 \angle 0^\circ \text{ V}$$

And,

$$20 \cos(4t - 30^\circ) \text{ V} = 20 \angle -30^\circ \text{ V}$$

Calculate the impedance of the 1 H inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(4)(1) \\ &= j4 \Omega \end{aligned}$$

Calculate the impedance of the 1 F capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(4)(1)} \\ &= \frac{-j}{4} \\ &= -j0.25 \Omega \end{aligned}$$

Step 2 of 5

Draw the circuit diagram in frequency domain.

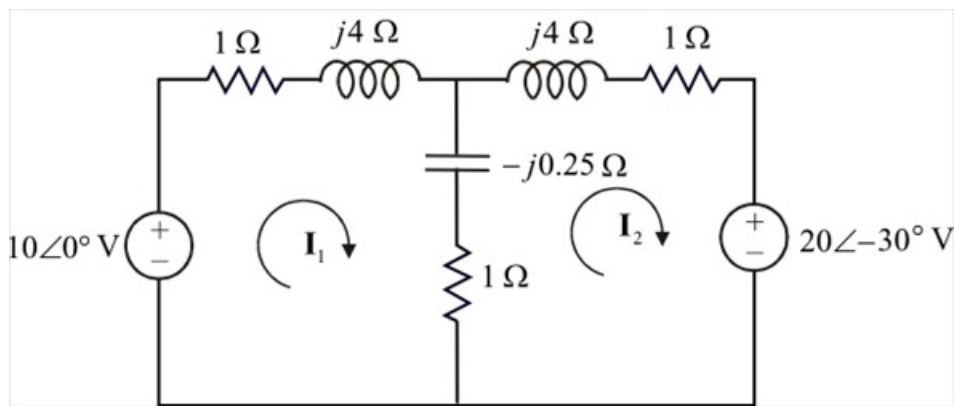


Figure 1

Step 3 of 5

Write the Kirchhoff's voltage law equation for loop 1.

$$-10\angle 0^\circ + (1 + j4)\mathbf{I}_1 + (1 - j0.25)(\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$(2 + j3.75)\mathbf{I}_1 - (1 - j0.25)\mathbf{I}_2 = 10 \quad \dots\dots (1)$$

Write the Kirchhoff's voltage law equation for loop 2.

$$(1 - j0.25)(\mathbf{I}_2 - \mathbf{I}_1) + (1 + j4)\mathbf{I}_2 + 20\angle -30^\circ = 0$$

$$-(1 - j0.25)\mathbf{I}_1 + (2 + j3.75)\mathbf{I}_2 = -20\angle -30^\circ$$

$$-(1 - j0.25)\mathbf{I}_1 + (2 + j3.75)\mathbf{I}_2 = -17.32 + j10 \quad \dots\dots (2)$$

Write the equations (1) and (2) matrix form.

$$\begin{bmatrix} 2 + j3.75 & -1 + j0.25 \\ -1 + j0.25 & 2 + j3.75 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -17.32 + j10 \end{bmatrix}$$

Step 4 of 5

Calculate the value of current, $\mathbf{I}_1$ using Cramer's rule.

$$\begin{aligned} \mathbf{I}_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{\begin{vmatrix} 10 & -1 + j0.25 \\ -17.32 + j10 & 2 + j3.75 \end{vmatrix}}{\begin{vmatrix} 2 + j3.75 & -1 + j0.25 \\ -1 + j0.25 & 2 + j3.75 \end{vmatrix}} \\ &= \frac{(10)(2 + j3.75) - (-1 + j0.25)(-17.32 + j10)}{(2 + j3.75)(2 + j3.75) - (-1 + j0.25)(-1 + j0.25)} \\ &= \frac{5.18 + j51.83}{-11 + j15.5} \\ &= \frac{5.18 + j51.83}{-11 + j15.5} \\ &= 2.07 - j1.8 \\ &= 2.74\angle -41^\circ \text{ A} \end{aligned}$$

Calculate the current in time domain.

$$i_1(t) = 2.74\cos(4t - 41^\circ) \text{ A}$$

Therefore, the mesh current, $i_1(t)$ is $2.74\cos(4t - 41^\circ) \text{ A}$.

Calculate the value of current, $\mathbf{I_2}$ using Cramer's rule.

$$\begin{aligned}
 \mathbf{I_2} &= \frac{\Delta_2}{\Delta} \\
 &= \frac{\begin{vmatrix} 2+j3.75 & 10 \\ -1+j0.25 & -17.32+j10 \end{vmatrix}}{\begin{vmatrix} 2+j3.75 & -1+j0.25 \\ -1+j0.25 & 2+j3.75 \end{vmatrix}} \\
 &= \frac{(2+j3.75)(-17.32+j10) - (10)(-1+j0.25)}{(2+j3.75)(2+j3.75) - (-1+j0.25)(-1+j0.25)} \\
 &= \frac{-62.14 - j47.45}{-11 + j15.5} \\
 &= -0.144 + j4.11 \\
 &= 4.11 \angle 92^\circ \text{ A}
 \end{aligned}$$

Calculate the current in time domain.

$$i_2(t) = 4.11 \cos(4t + 92^\circ) \text{ A}$$

Therefore, the mesh current, $i_2(t)$ is $4.11 \cos(4t + 92^\circ) \text{ A}$.